

Sydney Girls High School



Mathematics Extension 1

YEAR 12

2021 HSC Assessment Task 3

Reading time : 5 minutes

Writing time : 60 minutes

Total marks : 50

Topics: Further Calculus, Motion, Forces and Projectiles, Differential Equations and Mathematical Induction

General Instructions:

- There are four (4) questions which are not of equal value.
- Attempt all questions.
- Start each question on a new page.
- Write on one side of the paper only.
- Diagrams are NOT to scale, unless stated.
- Board-approved calculators may be used.
- Write using a black or blue pen.
- White-out or correction tape is NOT to be used.
- A Mathematics reference sheet is provided.
- Show all necessary working. Marks may be deducted for badly arranged work or incomplete working.
- A correct answer without necessary working will be awarded a maximum of 1 mark.

Student Name:_____ **Teacher Name:**_____

Question One (12 marks)

a) Differentiate the following with respect to x :

i) $y = \sin^{-1}(x^2)$ (2)

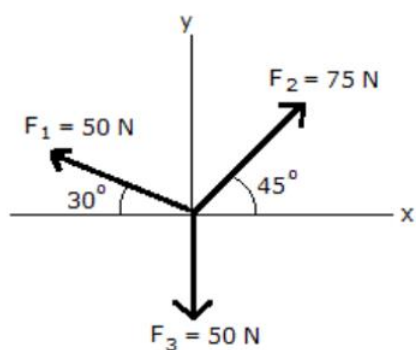
ii) $y = 2x \tan^{-1}(x)$ (2)

b) Find $\int_0^{0.125} \frac{2}{\sqrt{1-4x^2}} dx$. Write your answer correct to 3 decimal places. (2)

c) Solve $f'(x) = e^x + 6x - x^{\frac{1}{2}}$, given that $f(0) = 8$. (2)

d) Use the substitution $u = x^2$ to find $\int \frac{4x}{1+x^4} dx$. (2)

e) There are only three forces acting on an object as shown in the diagram. Determine the magnitude of the resultant force. (answer correct to 2 decimal places) (2)

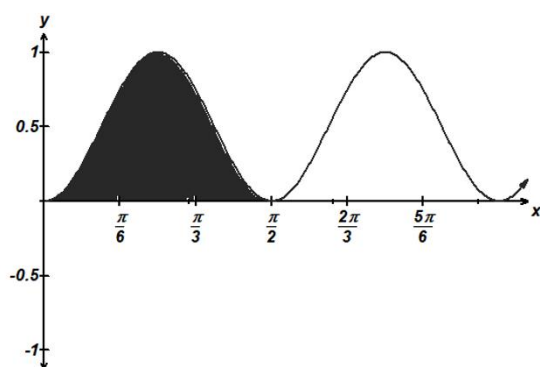


Question Two (13 marks)

- a) The graph of $y = 5\sqrt{1 - \frac{x^2}{4^2}}$ from $x = -2$ to $x = 2$ is rotated about the x -axis.
Evaluate the exact volume generated. (3)

- b) Find the general solutions of : $y' = x - xy$. (2)

- c) The diagram below shows the graph of $y = \sin^2 2x$. Find the shaded area. (3)



- d) Use the substitution $u = \frac{\pi}{x}$ to find $\int \frac{4 \cos\left(\frac{\pi}{x}\right)}{x^2} dx$. (2)

- e) Prove by induction for all positive integers of n .

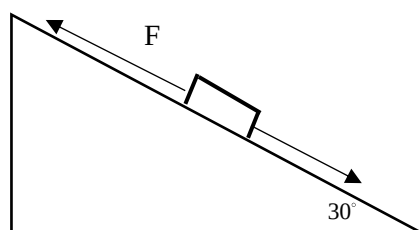
$$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \frac{1}{9 \times 13} + \dots + \frac{1}{(4n-3)(4n+1)} = \frac{n}{4n+1} \quad (3)$$

Question Three (13 marks)

a) Evaluate $\int_1^6 x\sqrt{x+3} \, dx$ using the substitution $u^2 = x+3$. (3)

b) A 12 kg block is sliding down a 30° incline with an acceleration of $1.5 \, \text{ms}^{-2}$.

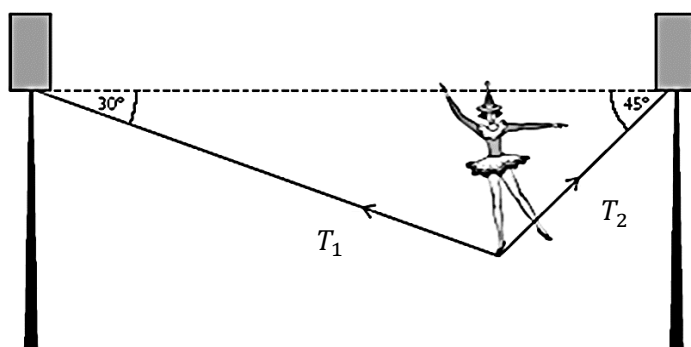
Calculate the magnitude of the frictional force, F . (Use $g = 10 \, \text{ms}^{-2}$) (2)



c) Prove by mathematical induction that $9^{n+2} - 4^n$ is divisible by 5, for all positive integers n . (3)

d) Given that $x \cos x$ is an odd function from $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$, evaluate $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (x - \cos x)^2 \, dx$. (2)

e) The diagram below shows a circus performer in a high wire act. The total mass of the performer is 76.5 kg. Acceleration due to gravity is given as $10 \, \text{ms}^{-2}$. Find the tensions T_1 and T_2 in the cable. (Answer correct to nearest whole number) (3)



Question Four (12 marks)

- a) The population, P , of animals in an environment in which there are scarce resources is increasing such that $\frac{dP}{dt} = P(100 - P)$, where t is time. When $t = 0$, $P = 10$.

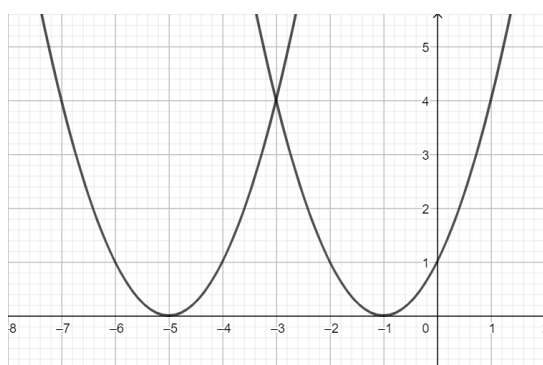
i) Show that $\frac{1}{100} \left(\frac{1}{P} + \frac{1}{100 - P} \right) = \frac{1}{P(100 - P)}$. (1)

ii) Find an expression for P in terms of t . (3)

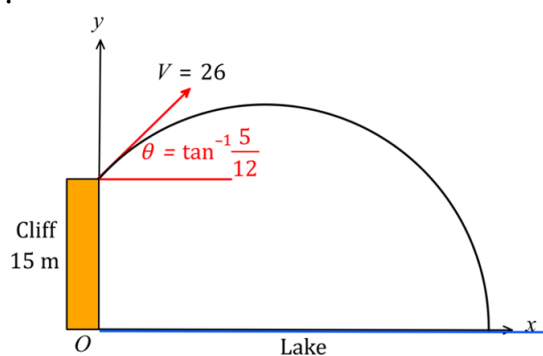
- b) In the diagram shown below, the area enclosed between two curves and the x -axis is rotated about the y -axis. Find the volume of the solid of revolution correct to one decimal place.

(Diagram is to scale)

(3)



- c) A rock is thrown from the top of a 15 m cliff with an initial velocity of 26 m/s and angle of projection of $\tan^{-1}\left(\frac{5}{12}\right)$ to the horizontal. The cliff overlooks a lake and the acceleration due to gravity is 10 ms^{-2} .



- i) Derive the equations of $y(t)$ and $x(t)$ in component form. (3)

- ii) Calculate the distance of the impact from the base of the cliff. (2)

THE END

Question One

$$\begin{aligned}\text{(a)(i)} \quad \frac{dy}{dx} &= \frac{1}{\sqrt{1-x^4}} \times 2x \\ &= \frac{2x}{\sqrt{1-x^4}}\end{aligned}$$

$$\text{(ii)} \quad \frac{dy}{dx} = \frac{2x}{1+x^2} + 2 \tan^{-1} x$$

$$\begin{aligned}\text{(b)} \quad \int_0^{0.125} \frac{2}{\sqrt{1-4x^2}} dx &= \left[\sin^{-1}(2x) \right]_0^{0.125} \\ &= \sin^{-1}(0.25) \\ &\approx 0.253\end{aligned}$$

The answer should have been in radians. No marks were lost for answer in degrees, this time.

$$\text{(c)} \quad f(x) = e^x + 3x^2 - \frac{2}{3}x^{\frac{3}{2}} + C$$

$$f(0) = e^0 + 3(0)^2 - \frac{2}{3}(0)^{\frac{3}{2}} + C$$

$$8 = 1 + C$$

$$C = 7$$

$$f(x) = e^x + 3x^2 - \frac{2}{3}x^{\frac{3}{2}} + 7$$

$$\begin{aligned}\text{(d)} \quad u &= x^2 & \int \frac{4x}{1+x^4} dx &= \int \frac{2}{1+u^2} du \\ \frac{du}{dx} &= 2x & &= 2 \tan^{-1}(u) + C \\ du &= 2x dx & &= 2 \tan^{-1}(x^2) + C\end{aligned}$$

$$\begin{aligned}
 \text{(e)} \quad F_1 &= -50 \cos 30^\circ \underline{i} + 50 \sin 30^\circ \underline{j} \\
 &= -25\sqrt{3}\underline{i} + 25\underline{j}
 \end{aligned}$$

$$\begin{aligned}
 F_2 &= 75 \cos 45^\circ \underline{i} + 75 \sin 45^\circ \underline{j} \\
 &= \frac{75}{\sqrt{2}}\underline{i} + \frac{75}{\sqrt{2}}\underline{j}
 \end{aligned}$$

$$F_3 = -50\underline{j}$$

$$F_1 + F_2 + F_3 = \left(\frac{75}{\sqrt{2}} - 25\sqrt{3} \right) \underline{i} + \left(25 + \frac{75}{\sqrt{2}} - 50 \right) \underline{j}$$

$$|F_1 + F_2 + F_3|^2 = \left(\frac{75}{\sqrt{2}} - 25\sqrt{3} \right)^2 + \left(\frac{75}{\sqrt{2}} - 25 \right)^2$$

$$|F_1 + F_2 + F_3| = 29.67 \text{ N}$$

Question Two

$$\begin{aligned} \text{(a)} \quad 2\pi \int_0^2 25 \left(1 - \frac{x^2}{4^2}\right) dx &= 50\pi \left[x - \frac{x^3}{3 \times 4^2} \right]_0^2 \\ &= 50\pi \left(2 - \frac{2 \times 4}{3 \times 4^2} \right) \\ &= 50\pi \left(2 - \frac{1}{6} \right) \\ &= \frac{275\pi}{3} \text{ units}^3 \end{aligned}$$

$$\text{(b)} \quad \frac{dy}{dx} = -x(y-1)$$

$$\int \frac{1}{y-1} dy = \int -x dx$$

$$\ln|y-1| = -\frac{x^2}{2} + C$$

Some student didn't not integrate the x side of the equation. Students should take care in the definition of the arbitrary constant.

$$|\pm(y-1)| = Ae^{-\frac{x^2}{2}} \quad \text{where } A = e^C > 0$$

$$y-1 = Ae^{-\frac{x^2}{2}} \quad \text{allow } A \neq 0$$

$$y = Ae^{-\frac{x^2}{2}} + 1 \quad \text{By removing all restrictions on } A \text{ this will include the constant solution } y = 1.$$

$$\begin{aligned}
 \text{(c)} \quad \int_0^{\frac{\pi}{2}} \sin^2(2x) \, dx &= \frac{1}{2} \int_0^{\frac{\pi}{2}} (1 - \cos(4x)) \, dx \\
 &= \frac{1}{2} \left[x - \frac{\sin(4x)}{4} \right]_0^{\frac{\pi}{2}} \\
 &= \frac{1}{2} \left(\frac{\pi}{2} - \frac{\sin(2\pi)}{4} \right) \\
 &= \frac{\pi}{4} \text{ units}^2
 \end{aligned}$$

Extension 1 students all should be able to integrate $\sin^2 x$ and $\cos^2 x$.

$$\begin{aligned}
 \text{(d)} \quad u &= \frac{\pi}{x} \\
 \frac{du}{dx} &= -\frac{\pi}{x^2} \\
 -\frac{1}{\pi} du &= \frac{1}{x^2} dx \\
 \int \frac{4 \cos\left(\frac{\pi}{x}\right)}{x^2} dx &= -\frac{4}{\pi} \int \cos(u) \, du \\
 &= -\frac{4}{\pi} \sin(u) + C \\
 &= -\frac{4}{\pi} \sin\left(\frac{\pi}{x}\right) + C
 \end{aligned}$$

Some students integrated u , to get $\pi \ln x$, instead of differentiating.

(e) Prove for $n = 1$

$$\begin{aligned} LHS &= \frac{1}{1 \times 5} & RHS &= \frac{1}{4(1) + 1} \\ &= \frac{1}{5} & &= \frac{1}{5} \\ & & &= LHS \end{aligned}$$

Assume for $n = k$

$$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \cdots + \frac{1}{(4k-3)(4k+1)} = \frac{k}{4k+1}$$

Prove for $n = k + 1$

$$\begin{aligned} &\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \cdots + \frac{1}{(4k-3)(4k+1)} + \frac{1}{(4(k+1)-3)(4(k+1)+1)} \\ &= \frac{k}{4k+1} + \frac{1}{(4k+1)(4k+5)} \\ &= \frac{k(4k+5)}{(4k+1)(4k+5)} + \frac{1}{(4k+1)(4k+5)} \\ &= \frac{4k^2 + 5k + 1}{(4k+1)(4k+5)} \\ &= \frac{(4k+1)(k+1)}{(4k+1)(4k+5)} \\ &= \frac{k+1}{4(k+1)+1} \end{aligned}$$

Therefore by the principles of mathematical induction
the statement is true for all positive integers of n .

Q3

a) $\int_1^6 x \sqrt{x+3} dx$

Let $u^2 = x+3 \therefore u = \sqrt{x+3}$

$x = u^2 - 3$

$\frac{dx}{du} = 2u \therefore dx = 2u du$

$x = 6 \rightarrow u = 3$

$x = 1 \rightarrow u = 2$

$$I = \int_2^3 (u^2 - 3) u \cdot 2u du$$

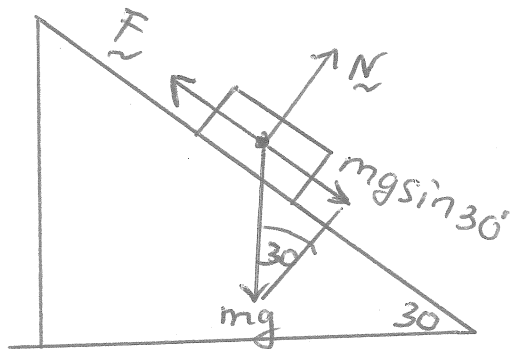
$$I = \int_2^3 (2u^4 - 6u^2) du$$

$$I = \left[\frac{2u^5}{5} - 2u^3 \right]_2^3$$

$$I = \frac{486}{5} - 54 - \left(\frac{64}{5} - 16 \right)$$

$$I = \frac{232}{5}$$

b)



$$ma = mg \sin 30^\circ - F$$

$$F = mg \sin 30^\circ - ma$$

$$F = 12 \times 10 \times \frac{1}{2} - 12 \times 1.5$$

$$|F| = 42 \text{ N}$$

Q3

c) prove it is true for $n = 1$

$$9^3 - 4 = 725 \text{ which is divisible by } 5$$

$\therefore$ It is true for $n = 1$

Assume it is true for $n = k$

$$9^{k+2} - 4^k = 5M$$

prove it is true for $n = k+1$

$$\begin{aligned} 9^{k+3} - 4^{k+1} &= 9(9^{k+2} - 4^k) + 5 \cdot 4^k \\ &= 9(5M) + 5 \cdot 4^k \\ &= 5(9M + 4^k) \end{aligned}$$

This is divisible by 5. Hence it is true for $n = k+1$. Proven by Mathematical Induction.

$$d) \int_{-\pi/2}^{\pi/2} (x - \cos x)^2 dx = 2 \int_0^{\pi/2} (x - \cos x)^2 dx$$

$$I = 2 \int_0^{\pi/2} (x^2 - 2x \cos x + \cos^2 x) dx$$

Note $\int_{-\pi/2}^{\pi/2} x \cos x dx = 0$ ($x \cos x$ is odd fn)

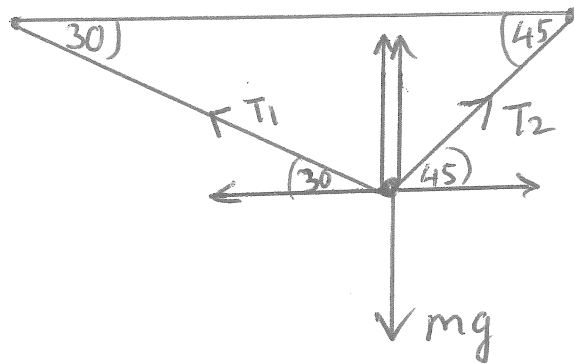
$$I = 2 \int_0^{\pi/2} \left(x^2 + \frac{1}{2} (\cos 2x + 1) \right) dx$$

$$I = \left[\frac{2x^3}{3} + \frac{1}{2} \sin 2x + x \right]_0^{\pi/2}$$

$$I = \frac{2}{3} \times \frac{\pi^3}{8} + 0 + \frac{\pi}{2} - 0 = \frac{\pi^3}{12} + \frac{\pi}{2}$$

Q4

e)



Horizontal: $T_1 \cos 30^\circ = T_2 \cos 45^\circ$ (1)

Vertical: $T_1 \sin 30^\circ + T_2 \sin 45^\circ = mg$ (2)

from (1) $T_1 = \frac{T_2 \cos 45^\circ}{\cos 30^\circ} = \frac{\sqrt{6}}{3} T_2$

Subs into (2)

$$\frac{\sqrt{6}}{3} T_2 + \frac{\sqrt{2}}{2} T_2 = 76.5 \times 10$$

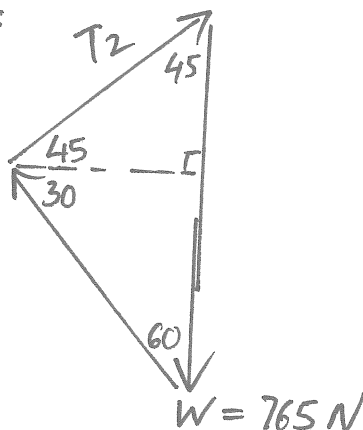
$$T_2 \left(\frac{\sqrt{6}}{3} + \frac{\sqrt{2}}{2} \right) = 765$$

$$T_2 = 685.9 \text{ (N)} = 686 \text{ N}$$

$$T_1 = \frac{\sqrt{6}}{3} \times 685.9 = 560 \text{ (N)}$$

Alternate Method:

Sine Rule



$$\frac{T_1}{\sin 45^\circ} = \frac{765}{\sin 75^\circ} \quad (1)$$

$$\frac{T_2}{\sin 60^\circ} = \frac{765}{\sin 75^\circ} \quad (2)$$

Q4. Task 3 Ext 1 2021

$$\begin{aligned} \text{ai) LHS} &= \frac{1}{100} \left(\frac{100 - P + P}{P(100 - P)} \right) \\ &= \frac{1}{100} \left(\frac{100}{P(100 - P)} \right) \\ &= \frac{1}{P(100 - P)} \\ &= \text{RHS} \end{aligned}$$

$$\text{ii) } \frac{dP}{dt} = P(100 - P)$$

$$\frac{dP}{P(100 - P)} = dt$$

$$\int \frac{1}{100} \left(\frac{1}{P} + \frac{1}{100 - P} \right) dP = \int dt$$

$$\int \frac{1}{P} + \frac{1}{100 - P} dP = \int 100 dt$$

$$\ln|P| - \ln|100 - P| = 100t + C$$

$$\ln \left| \frac{P}{100 - P} \right| = 100t + C$$

$$\left| \frac{P}{100 - P} \right| = e^{100t + C}$$

$$\left| \frac{P}{100 - P} \right| = e^C \cdot e^{100t}$$

$$\frac{P}{100 - P} = A e^{100t} \quad A \in \mathbb{R}$$

at $t=0$, $P=10$

$$\frac{10}{100 - 10} = A e^0 \quad \therefore A = \frac{1}{9}$$

* Many students didn't find the value of

A

$$\frac{P}{100-P} = \frac{1}{9} e^{100t}$$

$$\frac{9P}{100-P} = e^{100t}$$

$$9P = (100-P) e^{100t}$$

$$9P = 100 e^{100t} - P e^{100t}$$

$$9P + P e^{100t} = 100 e^{100t}$$

$$P(9 + e^{100t}) = 100 e^{100t}$$

$$P = \frac{100 e^{100t}}{9 + e^{100t}}$$

$$= \frac{\frac{100 e^{100t}}{e^{100t}}}{\frac{9}{e^{100t}} + \frac{e^{100t}}{e^{100t}}}$$

$$P = \frac{100}{9e^{-100t} + 1}$$

$$b) \quad y = (x+1)^2$$

$$y = (x+5)^2$$

$$y = (x+5)^2$$

$$x+5 = \pm\sqrt{y}$$

$$x+1 = \pm\sqrt{y}$$

$$x = -5 \pm \sqrt{y}$$

$$x = -1 \pm \sqrt{y}$$

$$V = \pi \int_0^4 (-5 + \sqrt{y})^2 - (-1 - \sqrt{y})^2 dy$$

$$= \pi \int_0^4 25 - 10\sqrt{y} + y - (1 + 2\sqrt{y} + y) dy$$

$$\pi \int_0^4 25 - 10\sqrt{y} + y - 1 - 2\sqrt{y} - y \, dy$$

$$\pi \int_0^4 24 - 12\sqrt{y} \, dy$$

$$12\pi \left[2y - \frac{2y^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^4$$

$$= 12\pi \left[8 - \frac{16}{\frac{3}{2}} \right]$$

$$= 32\pi \, \text{m}^3$$

* Many students couldn't

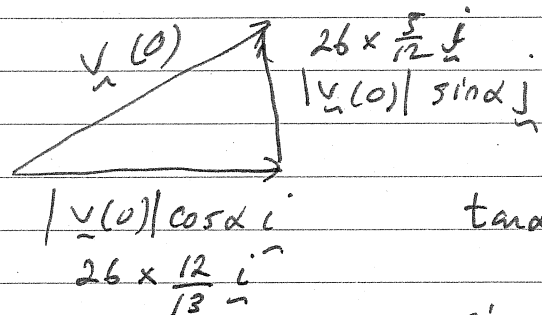
do this question

They needed to take the correct branch of the parabola

$$C(i) \quad \underline{a}(t) = -10 \underline{j}$$

$$\underline{v}(t) = \int -10 \underline{j} \, dt$$

$$= -10t \underline{j} + C$$



$$\tan \alpha = \frac{5}{12}$$

$$\sin \alpha = \frac{5}{13}$$

$$\text{at } t=0 \quad \underline{v}(t) = \underline{v}(0) \therefore C = \underline{v}(0)$$

$$\cos \alpha = \frac{12}{13}$$

$$\underline{v}(t) = |\underline{v}(0)| \cos \alpha \underline{i} + (|\underline{v}(0)| \sin \alpha - 10t) \underline{j}$$

$$= 26 \times \frac{12}{13} \underline{i} + \left(26 \times \frac{5}{12} - 10t \right) \underline{j}$$

$$= 24 \underline{i} + (10 - 10t) \underline{j}$$

$$\underline{r}(t) = 24t \underline{i} + (10t - 5t^2 + C) \underline{j}$$

$$t=0 \quad \underline{r}(t) = 15 \quad \therefore C = 15$$

$$\underline{v}(t) = 24 \underline{i} + (10t - 5t^2 + 15) \underline{j}$$

$$\text{ii) } y=0 \quad 10t - 5t^2 + 15 = 0 \quad \therefore (t-3)(t+1) = 0$$

$$t=3 \quad \text{as } t > 0 \quad \therefore x = 24 \times 3$$

$$= 72 \, \text{m}$$

Many students didn't show the correct steps.

Many stated the formulas rather than deriving.